

The Marker Protocol

Every experiment event the app emits is packed into one 64-bit integer and pushed to a single LSL outlet. This is the field layout, the value tables, and the rules a decoder needs to turn those integers back into trials.

STREAM NAME App Events	STREAM TYPE Markers	CHANNELS 1
FORMAT int64	RATE Irregular	SOURCE ID (empty)

§2 Bit layout

Seven fields fill the 64 bits. Every field is byte-aligned, and only `image_id` is wider than a single byte, so a decoder can read the sample as eight bytes rather than shifting bit by bit.

byte 7 block 63–56	byte 6 block_type 55–48	byte 5 trial 47–40	byte 4 state 39–32	bytes 3–2 image_id 31–16	byte 1 speed 15–8	byte 0 data 7–0
Field map, most significant byte first. Unused high bits within a field are always zero.						

Extraction				
FIELD	BITS	WIDTH	EXPRESSION	MEANING
block	63–56	8	<code>v >> 56 & 0xFF</code>	1-based block index; 0 outside an experiment
block_type	55–48	8	<code>v >> 48 & 0xFF</code>	Which kind of run this marker belongs to
trial	47–40	8	<code>v >> 40 & 0xFF</code>	1-based trial within the block; step number during calibration
state	39–32	8	<code>v >> 32 & 0xFF</code>	The phase within the trial
image_id	31–16	16	<code>v >> 16 & 0xFFFF</code>	Stimulus identifier — or the calibration result
speed	15–8	8	<code>v >> 8 & 0xFF</code>	Walking condition of the trial
data	7–0	8	<code>v & 0xFF</code>	Payload; currently only ratings

READ THE SAMPLE AS UNSIGNED

LSL delivers an *signed* int64. `block` occupies the top byte, so any block index of 128 or higher sets bit 63 and the sample arrives negative. Mask with `0xFFFF_FFFF_FFFF_FFFF` before shifting, or the block, block_type and trial fields will decode to nonsense. Realistic runs never reach block 128 — but a decoder that assumes so will fail silently rather than loudly.

§3 Field values

block_type

The most important field: it decides which of the others carry meaning. Three of the six values are not experiment blocks at all.

VALUE	NAME	MEANING
0	calibration	A speed calibration walk. Not experiment data.
1	stimulus	A trial that shows an image and is rated.
2	neutral	A trial that shows a fixation cross instead of an image, and is not rated.
3	pause	A break between blocks.
4	test	A practice run opened from the instructions. Never real data — exclude it.
5	session	The session-start marker. Carries no block and no trial.

state

Values 0–6 occur within trials; 7–11 occur only when `block_type` is `calibration`. The two ranges never overlap, so `state` means one thing regardless of context.

VALUE	NAME	EMITTED WHEN
0	None	A pause screen opens, or a session starts
1	Baseline	The fixation cross appears at the start of a trial
2	Stimulus	The stimulus phase begins
3	Go	The go signal — the participant starts walking
4	RatingPrompt	The response screen appears (on rated <i>and</i> unrated trials)
5	RatingValence	A valence rating is submitted; <code>data</code> holds it
6	RatingArousal	An arousal rating is submitted; <code>data</code> holds it
7	CalibrationStart	A calibration walk begins
8	CalibrationStop	A calibration walk ends
9	CalibrationDiscarded	The step was rejected and will be repeated
10	CalibrationConfirmed	The step was accepted
11	CalibrationResult	The calibration finished; <code>image_id</code> holds the result

speed

The walking condition assigned to the trial, fixed before the run starts and balanced within each block. Values are ordered slowest to fastest, so the field can be treated as an ordinal.

VALUE	NAME	TIME BUDGET
0	none	No walking in this marker's context
1	very_slow	1.2 ×
2	slow	1.1 ×
3	normal	1.0 ×
4	fast	0.9 ×
5	very_fast	0.8 ×

The multiplier scales the time allowed to walk the configured distance at the calibrated baseline speed. Conditions other than `normal` only appear if they were enabled in the settings for that run.

image_id

On a stimulus trial the 16 bits split into two parts:

BITS	PART	MEANING
15–14	quadrant	Valence/arousal category, 0–3
13–0	index	Position within that quadrant's image list

The quadrant is a two-bit pair: the high bit is valence, the low bit is arousal, each `0 = Low`, `1 = High`.

QUADRANT	FOLDER	VALENCE	AROUSAL
0	low_low	Low	Low
1	low_high	Low	High
2	high_low	High	Low
3	high_high	High	High

The index is the image's position in that folder's `.webp` files sorted by file name. That ordering is stable across machines and runs, but it is **relative to the image set**: adding or removing a file renumbers everything after it. The trial CSV records the file name next to each trial, which is how a recording is resolved back to actual images.

data

Currently only ratings, on the `RatingValence` and `RatingArousal` markers, as the integer 1–7 the participant pressed. Zero on every other marker. Because a rating is never 0, a zero here unambiguously means "no payload".

§4 Reading rules

Fields are not universally meaningful. `block_type` decides which to read; the rest are zero-filled rather than absent, and **zero is a legal value for several of them**, so a decoder must branch on `block_type` rather than test for zero.

Which fields carry meaning						
BLOCK_TYPE	BLOCK	TRIAL	STATE	IMAGE_ID	SPEED	DATA
session	0	0	None	—	—	—
calibration	0	step no.	7–11	result only	—	—
stimulus	1...	1...	1–6	image	condition	rating
neutral	1...	1...	1–4	—	condition	—
pause	1...	1	None	—	—	—
test	1...	1...	1–6	image	condition	rating

Three consequences worth stating explicitly:

- An `image_id` of 0 is a real image — quadrant `low_low`, index 0. It does not mean "no image". Only `block_type` tells you whether the field is an image at all.
- **test markers look exactly like stimulus markers** in every other field, including block and trial numbering. Filtering them out is a decoder's responsibility.
- Under `calibration`, `image_id` is not an image. On the `CalibrationResult` marker it holds the calibrated walking speed as $\text{km/h} \times 100$ — so `423` means 4.23 km/h. It is 0 on every other calibration marker.

§5 Marker sequences

Markers arrive in a fixed order within each unit of the run. Knowing the expected sequence is usually the fastest way to segment a recording.

A stimulus trial

1	Baseline — fixation cross appears
2	Stimulus — image shown
3	Go — go signal, walking starts
4	RatingPrompt — response screen appears
5	RatingValence — if valence rating is enabled
6	RatingArousal — if arousal rating is enabled

Both rating markers are configurable and either may be absent. If both are disabled the trial ends after **RatingPrompt**.

A neutral trial

1	Baseline
2	Stimulus — fixation cross, no image
3	Go
4	RatingPrompt — confirmation screen, nothing to rate

NOTE

A neutral trial still emits **Stimulus** and **RatingPrompt**. The phase structure is identical to a stimulus trial — what differs is that no image is shown and no rating markers follow. Do not use the presence of **RatingPrompt** to infer that a trial was rated.

A calibration step

1	CalibrationStart — walk begins
2	CalibrationStop — walk ends
3	CalibrationConfirmed — or CalibrationDiscarded

A discarded step repeats with the *same* `trial` number, so a step number appearing more than once is normal — the last occurrence is the accepted one. After the final step is confirmed, one `CalibrationResult` marker closes the calibration.

A whole session

Opening the experiment emits a single `session` marker. Then each entry of the trial plan runs in order, with `pause` markers where the plan has breaks. There is **no end-of-run marker** — the stream simply stops, so the end of a run must be inferred from the last trial or from the recording ending.

§6 Worked examples

Every value below is produced by the encoder itself, not written by hand.

BLOCK 1, TRIAL 4, FAST CONDITION, IMAGE HIGH_HIGH #127

MARKER	HEX	SIGNED INT64
Baseline	0x01010401C07F0400	72343474585666560
Stimulus	0x01010402C07F0400	72343478880633856
Go	0x01010403C07F0400	72343483175601152
RatingPrompt	0x01010404C07F0400	72343487470568448
RatingValence = 6	0x01010405C07F0406	72343491765535750
RatingArousal = 3	0x01010406C07F0403	72343496060503043

DECOMPOSING 0X01010406C07F0403

```

block      0x01 → 1
block_type 0x01 → stimulus
trial      0x04 → 4
state      0x06 → RatingArousal
image_id   0xC07F → quadrant 3 (high_high), index 127
speed      0x04 → fast
data       0x03 → arousal rating 3

```

0xC07F = 1100000001111111₂ – the top two bits are the quadrant, the remaining fourteen the index.

OTHER MARKERS

MARKER	HEX	DECODES TO
session start	0x0005000000000000	session, block 0, trial 0, None
pause, block 2	0x0203010000000000	pause, block 2, trial 1, None
neutral Go	0x0202010300000100	neutral, block 2, trial 1, Go, very_slow
calibration step 3 start	0x0000030700000000	calibration, step 3, CalibrationStart
calibration result	0x0000030B01A70000	calibration, CalibrationResult, 4.23 km/h